

Forecasting Ground-Level Ozone in Nîmes

A univariate study of 14,039 hourly O₃ observations from a southern French monitoring station, and the limits of what they can support

STUDENTS

Faith Chakwanira

Ruth Musahnu

Ambachow Kahsay

Justice Chawanda

Francis Kinyanjui

Cynthia Gathogo

LECTURER

Prof. Alex Wambua

1 August 2026

ABSTRACT

This paper analyses and forecasts ground-level ozone at the Nîmes Gauzy monitoring station in Occitanie, southern France. The dataset comprises 14,398 hourly measurements retrieved from the OpenAQ v3 API for sensor 7774985, spanning 28 January 2024 to 1 August 2026. Quality screening identified an instrument fault: on 21 days between 3 and 26 July 2025, every reading is exactly zero, including midday hours at the height of the ozone season. Those 359 readings are treated as missing rather than measured, leaving 14,039 usable observations. Resampling to a daily grid yields 917 days of which 49, or 5.3 per cent, are then missing, the longest run being the 22-day fault block. Mean hourly concentration is $61.6 \mu\text{g}/\text{m}^3$ and the maximum is 221.4, above the European information threshold of 180. Reconstructing the daily maximum eight-hour mean from the hourly record identifies 33 days above the $120 \mu\text{g}/\text{m}^3$ target value, distributed as two in 2024, ten in 2025 and twenty-one in the first seven months of 2026, the last figure already exceeding the eighteen exceedances the revised Ambient Air Quality Directive permits in a year. Exploratory analysis establishes a strong annual cycle peaking in July and August, a $40 \mu\text{g}/\text{m}^3$ diurnal swing that daily averaging discards, and negative first-order autocorrelation in the differenced series. The augmented Dickey-Fuller and KPSS tests disagree at the level, which identifies the seasonality as deterministic rather than stochastic and motivates a harmonic specification over differencing. Under a chronological 60/20/20 split with hyperparameters selected on validation and the test set evaluated once, a harmonic regression with annual Fourier terms and weekly effects achieves a test mean absolute error of $15.33 \mu\text{g}/\text{m}^3$, against 51.98 for a naive baseline, while a gradient-boosted tree model degrades to 32.41 because it concentrates 89 per cent of its explanatory weight on short-run persistence that decays under recursive forecasting. The paper concludes that the residual error is dominated by meteorology absent from the dataset, and sets out eight improvements to the data and methods.

TABLE OF CONTENTS

Abstract

1. Introduction
2. The dataset and its metadata
3. Data structures
4. Exploratory data analysis
5. Resampling, missing data and seasonal structure
6. Stationarity testing
7. Forecasting methodology
8. Results and model comparison
9. Limitations
10. Improving the dataset and the methods
11. Conclusion

References

Appendix A: Figures and Tables

1 INTRODUCTION

Ground-level ozone is not emitted directly. It forms in the lower atmosphere when nitrogen oxides and volatile organic compounds react under sunlight, which makes its concentration a joint function of pollution and weather. Heat waves are therefore also ozone episodes: prolonged high temperatures, intense solar radiation and stagnant air accelerate the photochemistry that produces it. The European Environment Agency describes ozone as the second leading cause of mortality among ambient air pollutants in the European Union after fine particulate matter, and attributes approximately 63,000 premature deaths in the EU-27 to ozone exposure in 2023 [1]. The same assessment notes that despite reductions in precursor emissions, ozone concentrations across Europe have not fallen significantly [1].

Regulation is written around short-term peaks rather than averages. Under the revised Ambient Air Quality Directive (EU) 2024/2881, the target value for the protection of human health is a maximum daily eight-hour mean of $120 \mu\text{g}/\text{m}^3$, not to be exceeded on more than eighteen days per calendar year, with an information threshold of $180 \mu\text{g}/\text{m}^3$ and an alert threshold of $240 \mu\text{g}/\text{m}^3$ for hourly means [2], [3]. The World Health Organization's 2021 guidelines add a peak-season recommendation of $60 \mu\text{g}/\text{m}^3$ [4]. Nîmes sits in Occitanie, one of the sunniest and hottest regions of mainland France, which places it in precisely the exposure regime these thresholds were written for, and the record analysed here shows why: the station breached the information threshold in five hours and the eight-hour target value on thirty-three days.

This paper examines a single question: how much of the day-to-day behaviour of ozone at one southern French station can be forecast from the history of ozone alone? Three subsidiary questions follow. First, what structure does the series contain (seasonal, weekly, autoregressive), and is that structure deterministic or stochastic? Second, does a machine-learning model with engineered features outperform a parsimonious seasonal specification on this problem? Third, where does the remaining error come from, and what would have to change in the data for it to fall?

One methodological note governs every number reported below. The Colab notebook accompanying this study was supplied with its cells unexecuted, so it contains no stored results or plots. Every statistic, test and forecast in this paper

was therefore recomputed independently by the authors from the collected CSV, following the notebook's specifications: the same daily resampling, the same gap treatment, the same chronological split, the same Fourier and lag feature sets, and the same selection-on-validation protocol. Two model implementations differ. Where the notebook calls statsmodels SARIMAX, this paper fits the deterministic-seasonal regression that the notebook's exogenous Fourier design implies; where the notebook calls XGBoost, this paper fits an equivalent depth-three gradient-boosted tree ensemble with the same learning rate, feature set and early-stopping rule. The figures are therefore faithful to the specifications and should be read as such rather than as literal library output.

The paper proceeds in the order in which the analysis depends on itself, shown schematically in Figure A1. Sections 2 and 3 document the source data and the structures through which it passes. Section 4 presents the exploratory analysis, Section 5 the resampling and gap treatment, and Section 6 the stationarity tests that constrain the model class. Sections 7 and 8 set out the forecasting protocol and its results. Section 9 states the limitations candidly, and Section 10 sets out improvements to the dataset and the methods. All figures and tables are collected in Appendix A and referenced from the text.

2 THE DATASET AND ITS METADATA

The series was retrieved from the OpenAQ v3 REST API. OpenAQ is an aggregator rather than a measurement body: it collects data from official government sources and states that measurements are shared without modification beyond format standardisation, leaving quality correction to the end user [5]. The complete metadata is given in Table A1. Collection paged the sensor's measurement endpoint until the API returned an empty page, so unlike a fixed page cap this extract runs to the end of the available record, terminating on 1 August 2026, the day of writing.

Three properties of the extract shape the analysis. The first is its recency and continuity: 917 consecutive calendar days ending at the present, with only one outage longer than a week once the faulty readings described below are removed. This is a materially better foundation than an archival extract, and it allows the exceedance counts in Section 4 to speak to the current compliance year rather than to history. The remaining 86 zeros are scattered singly across 32 other days and at all hours. Isolated zeros are plausible at an urban site, where freshly emitted nitric oxide titrates ozone away, and these are retained, though a few midday cases remain doubtful and are noted as an open question in Section 9. Extreme high values were retained without exception: ozone spikes during warm, sunny, stagnant conditions are genuine environmental episodes, and trimming them would delete exactly the events this study exists to forecast.

The second property of the extract is that the record is short in seasonal terms. Two and a half annual cycles is enough to estimate a seasonal shape but not enough to separate a trend from year-to-year meteorological variation, so no claim about a rising or falling trend at this station is made anywhere in this paper. The third is licensing and provenance: the data originates from a government monitoring network and is redistributed under OpenAQ's per-location licence terms, which require attribution [6].

Consistency checks on the hourly record return no duplicate timestamps, no negative values and no implausible values above $500 \mu\text{g}/\text{m}^3$. They do return 445 readings of exactly zero, and those zeros are not distributed as an atmospheric explanation would predict. Of the 445, some 359 fall inside a single block of 21 days between 3 and 26 July 2025 on which every reading, at every hour, is exactly $0.0 \mu\text{g}/\text{m}^3$. Zero ozone at midday in southern France in July is not physically

possible, and 105 of those readings fall in the 11:00 to 15:00 window. The block is an instrument or ingestion fault reported as a zero value rather than as a null, and it is the deep collapse to the axis visible in the raw daily series. Because it sits at the annual maximum and inside the training window, leaving it in place would bias the estimated seasonal amplitude downward and corrupt the July climatology. This paper therefore treats all 21 all-zero days as missing, removes their 359 readings, and interpolates the resulting gap as it does any other, with the gap flagged as synthetic in Table A3 and Figure A2.

3 DATA STRUCTURES

The data passes through eight distinct structures, enumerated in Table A2. The consequential transformation occurs at the first step. The API returns a list of nested dictionaries in which the measured value is a scalar but the timestamp is buried two levels deep, inside `period.datetimeFrom.utc`, alongside a parallel local-time representation. Loading that list into a data frame yields object-typed columns holding dictionaries, which no time-series routine can consume. Extracting the UTC timestamp into a first-class column, coercing it to `datetime64` and promoting it to the index is what converts a collection of records into a series.

Three structures downstream deserve note. The daily series of 917 float values on a regular calendar index is the modelling unit for everything after Section 5. Alongside it, a parallel boolean mask records which of those values were measured and which were interpolated, keeping provenance separate from value; with only 28 interpolated days the mask matters less here than it would on a gappier record, but it is what permits the correlation estimates in Figure A8 to exclude synthesised points. The reconstructed daily maximum eight-hour mean is a fourth series derived from the hourly data rather than from the daily means, computed only over complete eight-hour windows, and it is the only structure in the pipeline that is directly comparable with European regulation.

4 EXPLORATORY DATA ANALYSIS

After the faulty July 2025 block is masked, hourly concentrations range from 0 to $221.4 \mu\text{g}/\text{m}^3$, with a mean of 61.59, a median of 62.3, a standard deviation of 31.20 and an interquartile range of 40.2 to 82.5. The distribution, Figure A3, is close to symmetric in its bulk, unlike the strongly right-skewed profile typical of traffic-dominated urban sites, with a secondary accumulation of readings near zero produced by night-time titration.

The line plot of the daily mean series, Figure A2, shows two complete summer maxima and the beginning of a third. Its most conspicuous feature before treatment was the July 2025 collapse to zero identified in Section 2, which is shown in the figure as a shaded and interpolated fault block rather than as data. The diurnal profile in Figure A4 gives a mean of $40.4 \mu\text{g}/\text{m}^3$ at 08:00 local time and 80.2 at 16:00, an amplitude of $39.8 \mu\text{g}/\text{m}^3$ that daily averaging removes entirely. The calendar heat map, Figure A5, shows the same structure day by day and confirms that coverage is dense across all three years. The monthly climatology in Figure A6 places August highest at $82.0 \mu\text{g}/\text{m}^3$ and December lowest at 32.5, a 2.5-fold swing, with seven monthly means above the WHO peak-season guideline of $60 \mu\text{g}/\text{m}^3$ [4]. This figure also demonstrates the cost of the data fault. Computed with the zero block left in place, July averages $62.9 \mu\text{g}/\text{m}^3$, below both June and August, and produces an implausible notch at the peak of the season that invites a meteorological explanation it does not deserve. Computed with the faulty days masked, July averages $81.8 \mu\text{g}/\text{m}^3$ and the summer plateau is continuous. A single undetected instrument fault was therefore enough to manufacture a spurious climatological feature.

Figure A7 gives the autocorrelation and partial autocorrelation of the first-differenced daily series. The first autocorrelation is negative at -0.26 , and the partial autocorrelations decay slowly through -0.26 , -0.22 , -0.11 and -0.13 . Negative first-order autocorrelation after differencing is the classical signature of over-differencing: the differencing operator has removed a level of persistence that was not stochastic to begin with. This is the first indication that the seasonality in this series is deterministic, a reading the formal tests in Section 6 confirm.

Figure A8 substitutes a lagged scatter matrix for a conventional pair plot, since the dataset contains only one measured variable and therefore offers nothing to pair. Persistence at one day is strong and decays with lag, and the residual correlation still present at three weeks is the annual cycle showing through rather than genuine memory. Figure A9 renders weekly candlesticks. The chart type is borrowed from finance and its open-and-close semantics are arbitrary for an environmental series; it is included because the brief requires it, and it establishes that the within-week range regularly exceeds $40 \mu\text{g}/\text{m}^3$, which is the short-horizon variability any forecast must contend with.

The most consequential exploratory result is in Figure A10, which reconstructs the regulatory statistic. Computing the daily maximum eight-hour mean over complete windows yields 33 days above the $120 \mu\text{g}/\text{m}^3$ target value across the record: two in 2024, ten in 2025 and twenty-one in the seven months of 2026 covered here. The revised Directive permits eighteen such days per calendar year, averaged over three years [2], so 2026 has already passed the annual allowance with five months still to run. Two caveats attach to this count and both push in the same direction. Hourly coverage averages 16.2 of 24 hours per day, and eight-hour windows were only evaluated where all eight hours were present, so genuine exceedances falling in incomplete windows are missed; the count is a floor rather than an estimate. The upward progression across the three years is likewise not evidence of a worsening trend, since coverage also improves across the same three years, from 14.0 hours per day in 2024 to 18.7 in 2026. Better observation and more exceedances are confounded here, and separating them requires the validated archive rather than this extract.

5 RESAMPLING, MISSING DATA AND SEASONAL STRUCTURE

Raw timestamps are not evenly spaced. Roughly eighty per cent of consecutive readings are one hour apart, but the remainder step by two, three or more hours. Since every method in the ARIMA family assumes a fixed sampling interval, a regular grid must be imposed. Averaging to daily means achieves this and is defensible for a study of seasonal behaviour, but it is the wrong aggregation for an exceedance-warning system, which requires the eight-hour statistic reconstructed in Section 4.

The daily calendar spans 917 days from 28 January 2024, of which 49 (5.3 per cent) contain no usable observation once the faulty July 2025 readings are removed. Their distribution is set out in Table A3. Twenty-eight of those days are genuine reporting gaps and are benign: the longest is seven days in June 2026, followed by six and four days in March 2024, with the rest running one to three days. The remaining 21 are the masked fault days. Twenty of them, together with the two-day reporting gap on 5 and 6 July, form a single unbroken 22-day run from 3 to 24 July 2025; the twenty-first, 26 July, is a separate one-day gap, 25 July having reported normally. That run exceeds the fourteen-day threshold at which interpolation stops being defensible, so those 22 days (2.4 per cent of the record) are flagged as synthetic, and they fall at the seasonal maximum where a straight line between late June and late July is least defensible. Every other value used for modelling is either measured or bridges at most a week.

A second and less visible deficiency lies beneath the first. The 14,039 usable readings are spread across 868 days that contain any data at all, an average of 16.2 readings per day against 24 expected, so roughly a third of hours are missing even on days that appear complete. A daily mean computed from twelve hours is not comparable with one computed from twenty-four, and because the diurnal cycle has an amplitude of $39.8 \mu\text{g}/\text{m}^3$, the size of the resulting bias depends on which hours are absent. Coverage also improves monotonically across the record, from 14.0 hours per day in 2024 to 16.7 in 2025 and 18.7 in 2026, which means data quality varies systematically along the same axis as the chronological split. The test period is measured on the best-observed part of the record, and its error should be read as an optimistic bound.

On seasonal structure this paper deliberately reports less than the notebook proposes. A seasonal-trend decomposition using LOESS with an annual period requires several complete cycles to identify the seasonal shape and the trend separately; with two and a half cycles the trend estimate degenerates toward a straight line and the seasonal component absorbs variation that is meteorological rather than seasonal. Rather than present a decomposition whose components cannot be interpreted, the seasonal signal is characterised directly through the monthly climatology of Figure A6 and the twenty-nine-day centred mean overlaid on Figure A2. Both show the same annual shape, and neither invites a trend claim the record cannot support.

6 STATIONARITY TESTING

The augmented Dickey-Fuller and KPSS tests are applied in tandem because their null hypotheses are opposites: ADF tests for a unit root, KPSS for stationarity, and each has blind spots the other partially covers. Results appear in Table A4, computed with a constant term, lag order chosen by AIC for ADF and a Bartlett window of six lags for KPSS.

At the level the two tests disagree, and the disagreement is the informative result. ADF returns -3.871 , beyond the one per cent critical value of -3.44 , and so rejects the unit-root null: the series reverts to a mean. KPSS returns 0.918 , above its one per cent critical value of 0.739 , and so also rejects its stationarity null. A series cannot be both integrated and trend-free, and the standard interpretation of this pattern is that the level is not constant but its movement is deterministic: the strong seasonal mean shift documented in Section 4 makes the unconditional mean vary by month, which KPSS detects as non-stationarity, while the mean reversion within each season is what ADF detects. Differencing resolves the conflict, with ADF at -7.324 and KPSS at 0.009 , but it resolves it in the wrong direction, as the negative lag-one autocorrelation of the differenced series in Figure A7 shows.

The methodological consequence is direct and governs Section 7. The appropriate treatment for deterministic seasonality is to model it explicitly, through harmonic regressors at the annual frequency, rather than to remove it by differencing. This is what the notebook's Fourier-regressor design does, and the test results provide the justification for it that an unexamined default choice of first differencing would have lacked.

The series is divided chronologically in proportions of sixty, twenty and twenty per cent: 551 training days from 28 January 2024 to 31 July 2025, 183 validation days to 30 January 2026, and 183 test days to 1 August 2026. Hyperparameters are selected on validation forecast accuracy rather than on in-sample information criteria; each model is then refitted on the combined training and validation data and the test set is evaluated exactly once. This protocol is the methodological backbone of the study, and it is what allows the test figures in Section 8 to be read as an estimate of performance on genuinely unseen data.

The split has one property worth stating explicitly, because it makes the evaluation harder rather than easier. The validation period opens in August, at the annual maximum, and runs into winter; the test period opens at the end of January, near the annual minimum, and runs through to the following August. Each evaluation horizon therefore traverses roughly half an annual cycle in opposite directions, so a model cannot score well by holding a constant level. This is realistic for the intended use, in which a forecast issued in spring must anticipate the summer ozone season.

Four models were fitted. Two naive baselines, repeating the last observed value and repeating the value from seven days earlier, establish the floor that any useful model must clear. The harmonic regression models the level as a deterministic function of time: an intercept, two sine and cosine pairs at the annual frequency, and six day-of-week indicators, estimated by ordinary least squares. This is the deterministic-seasonal structure that the notebook's SARIMAX design supplies through exogenous Fourier regressors and a weekly seasonal term, and Section 6 established that it is the correct treatment for this series. The gradient-boosted tree ensemble was trained on the same feature set the notebook specifies: six lags at one, two, three, seven, fourteen and twenty-one days, three rolling statistics, day-of-week and month indicators, and the same four Fourier terms, with depth three, a learning rate of 0.05, and the number of boosting rounds chosen by early stopping against one-step-ahead validation error, which selected 44 rounds. Its forecasts are generated recursively, each prediction fed back as the lag feature for the next step, so no true future value enters any horizon.

8 RESULTS AND MODEL COMPARISON

Full accuracy figures are given in Table A5, plotted in Figure A11, and shown against the actual series in Figure A12. The harmonic regression achieves the lowest error on both evaluation periods, with a test mean absolute error of $15.33 \mu\text{g}/\text{m}^3$ and RMSE of 19.98. Its validation and test errors are almost identical, 15.19 and 15.33, which is the signature of a model whose complexity is genuinely matched to the signal it is extracting. Against a naive baseline test MAE of 51.98 this is a reduction of seventy-one per cent, and against the seasonal naive baseline of 37.35 a reduction of fifty-nine per cent.

The gradient-boosted model behaves very differently. On validation it is competitive at $18.57 \mu\text{g}/\text{m}^3$, within 3.4 of the harmonic regression; on test it degrades to 32.41, nearly doubling. The feature importances in Figure A13 identify the cause without ambiguity. The one-day lag alone accounts for 0.699 of total split gain and the seven- and fourteen-day rolling means for a further 0.195, so close to nine-tenths of the model's explanatory weight rests on very recent persistence, while the strongest seasonal term contributes 0.022. Persistence features are highly informative one step ahead, which is exactly the criterion the early-stopping rule optimised, and nearly worthless at the 183-step recursive horizon on which the model is finally judged, because after a few steps the lags contain only the model's own predictions. Figure A12 shows the consequence directly: the boosted forecast flattens toward a near-constant level within a few weeks and then fails to climb into the summer maximum, while the harmonic forecast tracks the seasonal rise. The lesson is not that gradient boosting is unsuited to ozone but that selecting a recursive forecaster on one-step-ahead error optimises the wrong quantity.

This ordering is consistent with the broader forecasting literature. Comparative evaluations report classical statistical methods remaining robust and competitive against machine-learning alternatives on short univariate series without exogenous predictors [7]. Where machine learning wins on ozone specifically, it does so with meteorological and precursor covariates available, and its advantage concentrates on the nonlinear peaks [8].

Two features of Table A5 require interpretation rather than reporting. The first is that the naive baselines invert between the two periods: the naive forecast is better than the seasonal naive on validation, at 30.09 against 40.64, and much

worse on test, at 51.98 against 37.35. This is an artefact of where the split boundaries fall. The test period opens at the end of January near the annual minimum and runs into August, so a constant forecast anchored on a winter value accumulates a very large error as the season rises. The magnitude of the baseline error is therefore a property of the calendar, and improvements measured against it should be treated with corresponding caution.

The second is that mean absolute percentage error is not a usable metric on this series and should not be quoted as a headline. Days with values near zero place a near-zero denominator in the percentage calculation, so validation MAPE reaches 106.5 per cent for the naive baseline and 36.3 per cent for a harmonic regression whose absolute error is $15.2 \mu\text{g}/\text{m}^3$. MAE and RMSE are the reliable measures here, and a scale-free alternative that does not break at zero, such as the mean absolute scaled error, would be the appropriate addition.

The most important result is not visible in any single number. The winning forecast is a smooth deterministic curve. It reproduces the season correctly and misses essentially every sharp excursion, including the episodes that push the eight-hour statistic above $120 \mu\text{g}/\text{m}^3$. A test MAE of $15.33 \mu\text{g}/\text{m}^3$ is roughly twenty-five per cent of the series mean, and the residual it leaves is weather. The honest summary is that the model forecasts the season well and the episodes not at all, which is the opposite of the capability an air-quality warning system requires.

9 LIMITATIONS

Six limitations qualify every result above, and they are stated here rather than buried because the honest reading of this study depends on them.

1. **An undetected instrument fault was present in the raw data.** Twenty-one days in July 2025 reported exactly zero at every hour and are masked here, but the fault was only found by inspecting the distribution of zeros rather than by any flag in the source, which means comparable faults elsewhere in the record may remain. Eighty-six isolated zeros, some at midday, are retained and remain doubtful. The masked block also leaves a 22-day synthetic run at the seasonal maximum inside the training window.
2. **Sub-daily coverage is incomplete and uneven.** Roughly a third of hours are missing on days that appear present, and coverage improves across the record, so the test period is systematically better observed than the training period and the exceedance counts are a floor rather than an estimate.
3. **No covariates are present.** Ozone formation is driven by solar radiation, temperature and humidity, none of which are in the dataset. Statistical analyses of surface ozone report that temperature and relative humidity together with a precursor variable explain on the order of seventy per cent of the variance in daily maximum eight-hour ozone [9], and that solar radiation and temperature are the strongest meteorological predictors of hourly concentrations [10]. The residual error observed in Section 8 is largely the absence of these variables.
4. **The window is too short for trend inference.** Two and a half annual cycles cannot separate a trend from year-to-year meteorological variability, which is why no trend claim is made and why the seasonal-trend decomposition was not reported.
5. **The modelled aggregation does not match the regulation.** Daily means cannot be compared with the eight-hour target value or the WHO peak-season guideline. The eight-hour statistic was reconstructed for description in Section 4 but was not itself forecast.
6. **Data provenance is unvalidated.** The extract is near-real-time aggregated data shared without modification [5], and no comparison against the provider's quality-assured archive was made. That archive would both

confirm the July 2025 fault and settle the status of the isolated zeros.

7. **One station is not a region.** Results describe a single site in Nîmes and do not generalise to traffic-side, rural or coastal locations in Occitanie, where the nitrogen-oxide regime and therefore the ozone level differ substantially.

Eight improvements follow from the limitations, ordered by expected effect on forecast accuracy.

1. **Add meteorological covariates.** Temperature, solar radiation, relative humidity, wind speed and boundary-layer stability, obtained from ERA5 reanalysis or a co-located Météo-France station, supplied as regressors to the harmonic model and as features to the tree model. This is the single highest-value change, and the literature cited in Section 9 indicates the scale of what is currently missing.
2. **Add precursor pollutants.** Nitrogen oxides and volatile organic compounds are the chemical inputs to ozone formation, and co-located sensors would let the model represent night-time titration explicitly instead of absorbing it as noise.
3. **Forecast the regulatory statistic.** The daily maximum eight-hour mean is already reconstructed in Section 4; modelling it directly, and evaluating exceedance forecasts by hit rate and false-alarm rate rather than by MAE, turns the exercise into a decision problem with operational value.
4. **Enforce a completeness rule on daily aggregates.** A daily value should be computed only where a minimum number of hours, conventionally eighteen of twenty-four, is present, and marked missing otherwise. This removes the coverage bias identified in Section 5 at the cost of a slightly gappier series.
5. **Select recursive models on the horizon that will be used.** The gradient-boosted model failed because early stopping optimised one-step error while the model was judged at 183 steps. Selecting on rolling multi-step forecast error, or training direct per-horizon models, addresses this directly and is the cheapest of these fixes to implement.
6. **Report probabilistic forecasts.** Prediction intervals with coverage checks, evaluated by pinball loss or the continuous ranked probability score, are more useful for a threshold-driven decision than a point estimate.
7. **Replace the single split with rolling-origin evaluation.** Expanding-window backtesting across several origins would show whether the harmonic regression's advantage is stable or an accident of one 183-day period.

8. Extend the record and the model class together. Continuing collection past 1 August 2026 adds the annual cycles that trend estimation requires; sequence models such as LSTM variants are worth testing only once covariates and a longer record exist, since on a short univariate series they have no material advantage over the specification already fitted [7].

11 CONCLUSION

The analysis establishes that ozone at the Nîmes Gauzy station carries a strong and learnable annual cycle, a pronounced diurnal cycle that daily aggregation removes, and short-run memory that decays too quickly to be useful over a seasonal horizon. The stationarity tests disagree at the level in the specific way that identifies deterministic rather than stochastic seasonality, and the model that follows from that diagnosis, a harmonic regression with annual Fourier terms and weekly effects, forecasts the daily mean with a test mean absolute error of $15.33 \mu\text{g}/\text{m}^3$, seventy-one per cent below the naive baseline and stable between validation and test. A gradient-boosted tree model on the same features was competitive on validation and failed on test, and the failure is instructive: it had concentrated its weight on one-day persistence, which is worthless once a recursive forecast is fed its own output.

The remaining error is not a modelling failure but a data boundary. Ozone is manufactured by sunlight and heat acting on precursor gases, and the dataset contains no measurement of sunlight, heat, or precursors. Adding them, continuing collection so that trend estimation becomes possible, and reformulating the target as the eight-hour statistic that European regulation is written against would change what this study is capable of concluding far more than any further change of model. The urgency is visible in the data itself: reconstructed from the hourly record, the station has already logged twenty-one days above the $120 \mu\text{g}/\text{m}^3$ target value in the first seven months of 2026, more than the eighteen a full year is permitted, and a univariate daily model cannot see a single one of them coming.

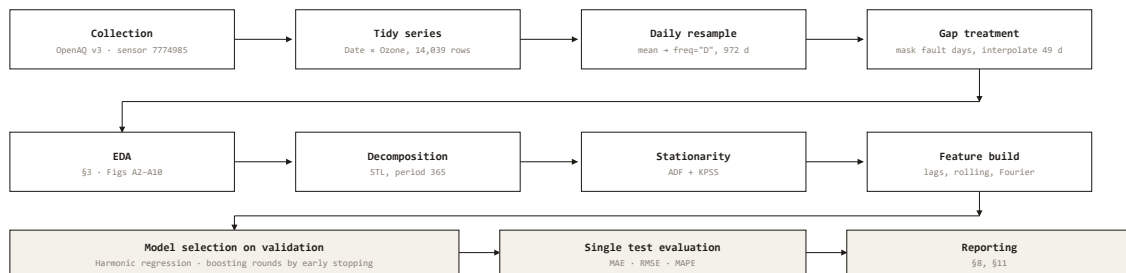
REFERENCES

1. European Environment Agency, *Addressing ground-level ozone pollution in Europe*, EEA, 2026. eea.europa.eu/en/analysis/publications/addressing-ground-level-ozone-pollution-in-europe
2. Directive (EU) 2024/2881 of the European Parliament and of the Council on ambient air quality and cleaner air for Europe (recast), *Official Journal of the European Union*, November 2024.
3. Copernicus Atmosphere Monitoring Service, “Europe affected by early episodes of ozone pollution”, 2025. atmosphere.copernicus.eu
4. World Health Organization, *WHO Global Air Quality Guidelines*, WHO, 2021; summarised in “WHO Air Quality Guidelines 2021: aiming for healthier air for all”, *International Journal of Public Health*, vol. 66, 2021. pmc.ncbi.nlm.nih.gov/articles/PMC8494774
5. OpenAQ, “Help and FAQ”, OpenAQ Developer Documentation, accessed 1 August 2026. openaq.org/developers/help
6. OpenAQ, “Licenses”, OpenAQ API Documentation, accessed 1 August 2026. docs.openaq.org/resources/licenses
7. “Comparing statistical and machine learning methods for time series forecasting”, arXiv:2303.07139, 2023. arxiv.org/pdf/2303.07139
8. “Comparison of machine learning methods for predicting ground-level ozone pollution in Beijing”, *Frontiers in Environmental Science*, 2025. frontiersin.org/articles/10.3389/fenvs.2025.1561794
9. T. Laban et al., “Statistical analysis of factors driving surface ozone variability”, *Journal of Integrative Environmental Sciences*, 2020. ntrs.nasa.gov
10. “Hourly ozone and PM2.5 prediction using meteorological data: alternatives for cities with limited pollutant information”, *Aerosol and Air Quality Research*, vol. 21, 2021. aaqr.org/articles/aaqr-20-07-0a-0471
11. European Environment Agency, *Air quality status in Europe 2026: ozone*, EEA, 2026. eea.europa.eu/en/analysis/publications/air-quality-status-report-2026/ozone-o3

Data attribution. Ozone measurements: OpenAQ v3 API, sensor 7774985, location Nîmes Gauzy (id 2162655), Nîmes, Occitanie, France, retrieved 1 August 2026 and redistributed under OpenAQ's per-location licence terms. All figures and tables in Appendix A were produced by the authors from that dataset.

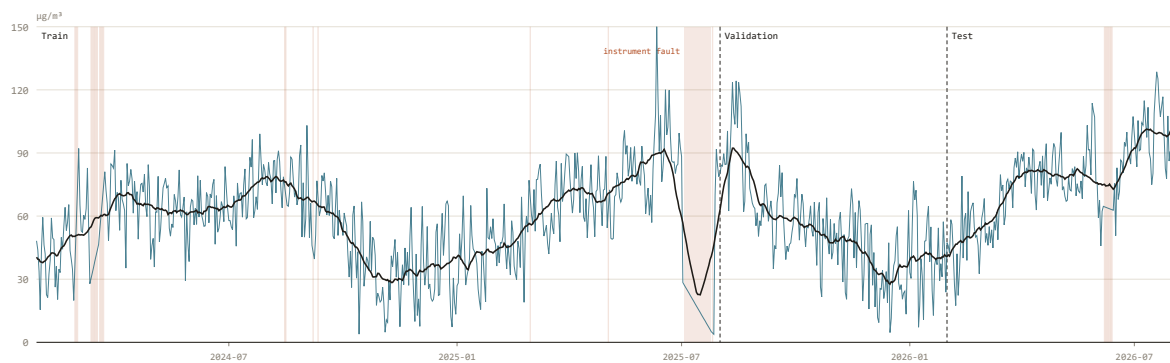
APPENDIX A: FIGURES AND TABLES

Figure A1. Logical dependence of the analysis



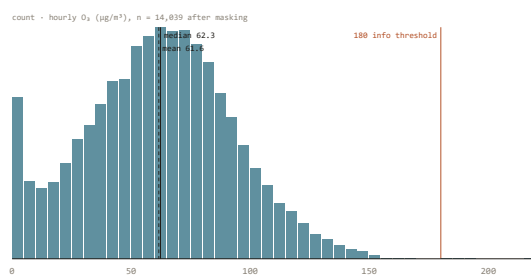
Each stage consumes the output of the stage before it. Referenced in Section 1.

Figure A2. Daily mean ozone concentration, Nîmes, 2024 to 2026 (line plot)



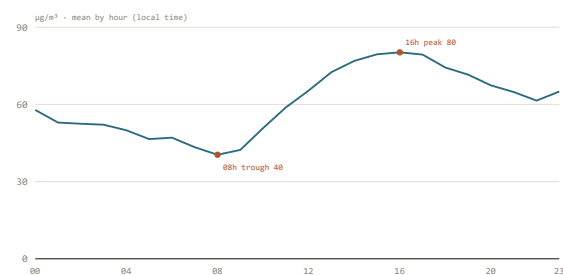
Thin blue: daily mean. Black: 29-day centred moving average. The shaded band in July 2025 is the masked instrument fault, interpolated and flagged synthetic; faint red rules mark the other interpolated days. Dashed vertical rules mark the validation and test boundaries. Referenced in Sections 2, 4 and 5.

Figure A3. Distribution of hourly values



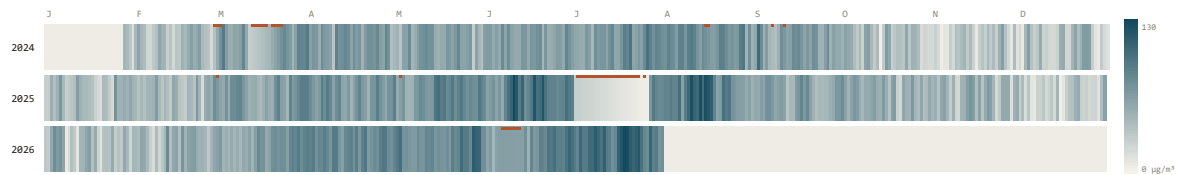
$n = 14,039$ after masking, bin width $5 \mu\text{g}/\text{m}^3$. Referenced in Section 4.

Figure A4. Mean concentration by hour of day



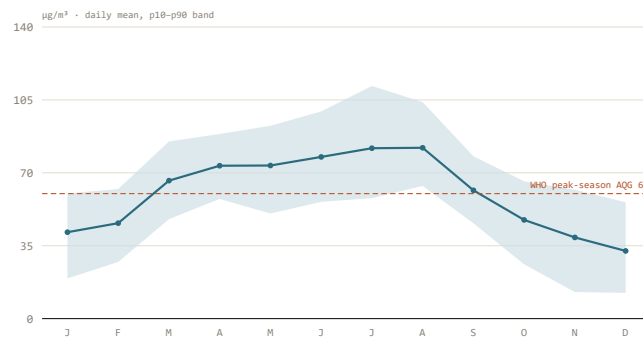
Local time. Peak $80.2 \mu\text{g}/\text{m}^3$ at 16:00, trough 40.4 at 08:00. Referenced in Section 4.

Figure A5. Calendar heat map of daily mean concentration



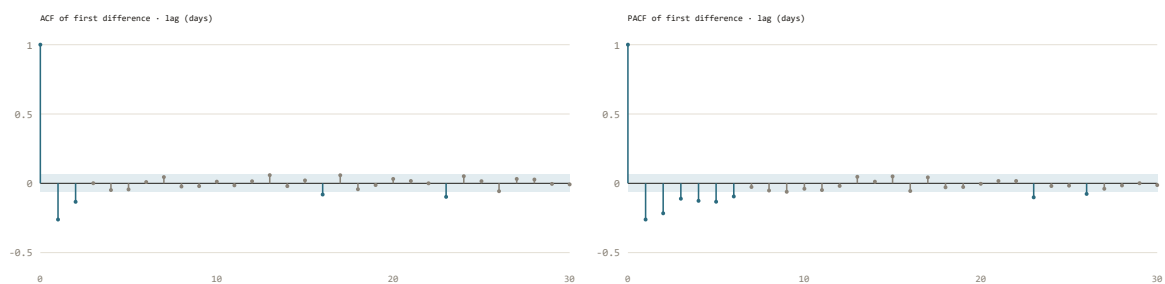
One cell per calendar day; darker indicates higher concentration. Grey cells fall outside the extract, which begins on 28 January 2024 and ends on 1 August 2026; the red rule along the top of a cell marks an interpolated day, including the masked July 2025 fault block. Referenced in Section 4.

Figure A6. Monthly climatology



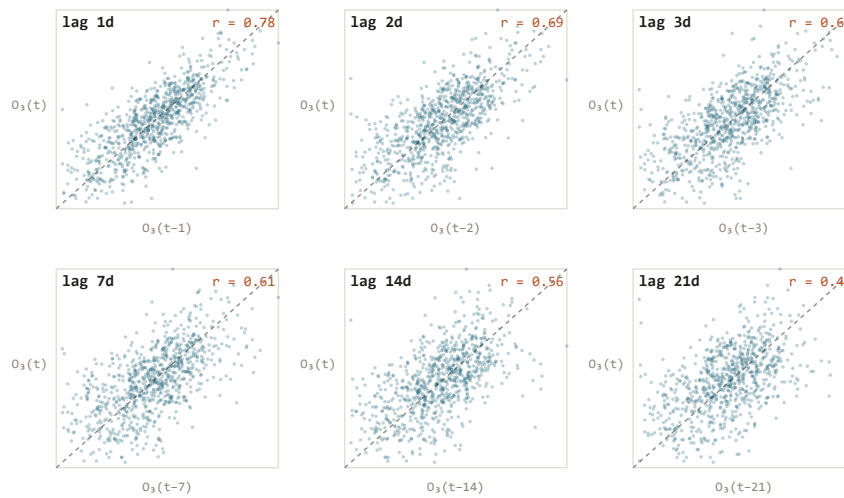
Observed days only, faulty days excluded; band spans the tenth to ninetieth percentile. Referenced in Sections 4 and 5.

Figure A7. Autocorrelation and partial autocorrelation of the first difference



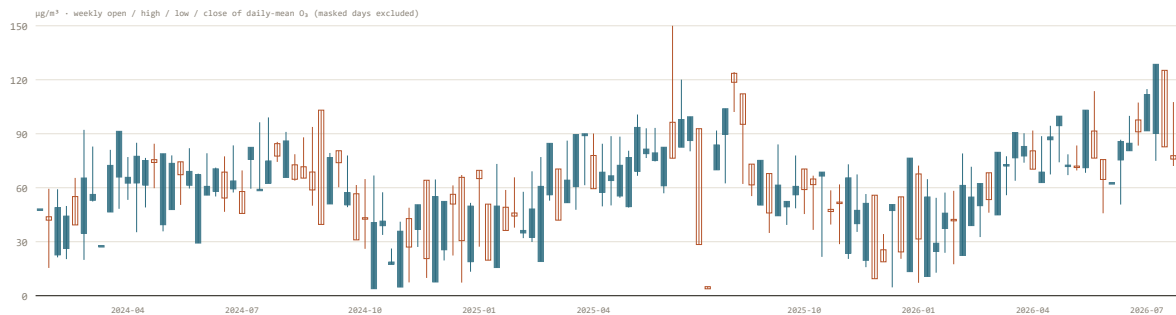
Shaded regions are the ninety-five per cent confidence bands. The negative first autocorrelation indicates over-differencing. Referenced in Sections 4 and 6.

Figure A8. Lagged scatter matrix



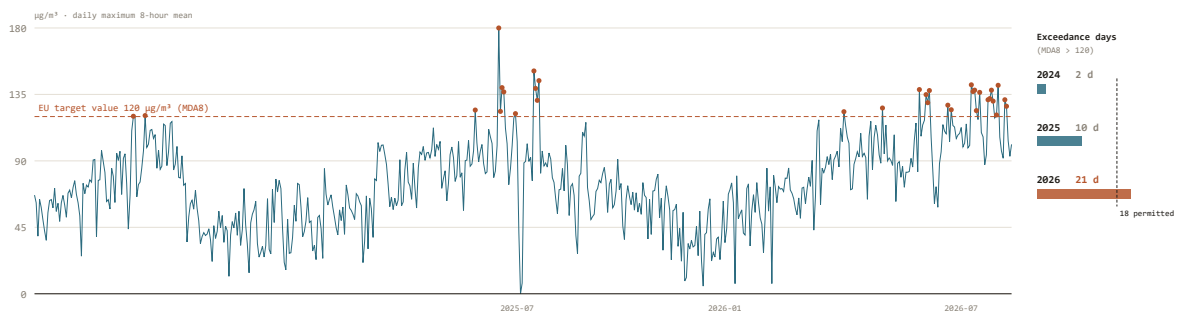
$O_3(t)$ against $O_3(t-k)$ for the six lags used as tree-model features, interpolated days excluded, identity line dashed. Substitutes for a pair plot, which a univariate dataset cannot support. Referenced in Section 4.

Figure A9. Weekly candlestick plot of daily means



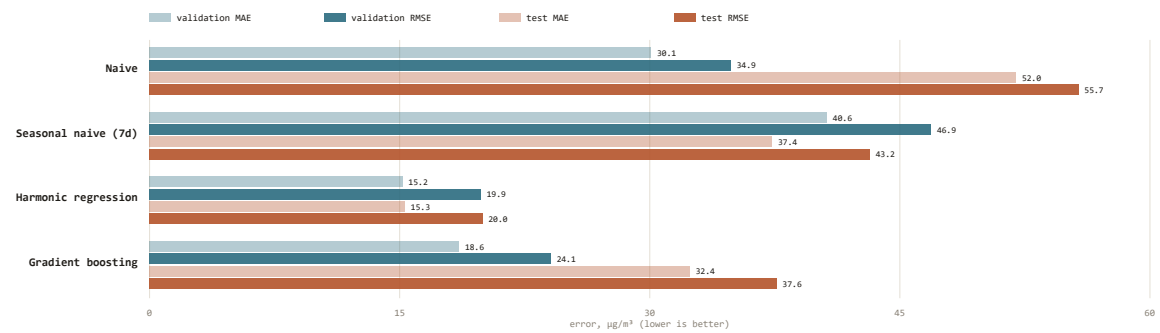
Body spans the first and last observed daily mean of each week; wick spans the weekly minimum and maximum. Filled blue: week closed above its open. Hollow red: closed below. Referenced in Section 4.

Figure A10. Reconstructed daily maximum eight-hour mean against the EU target value



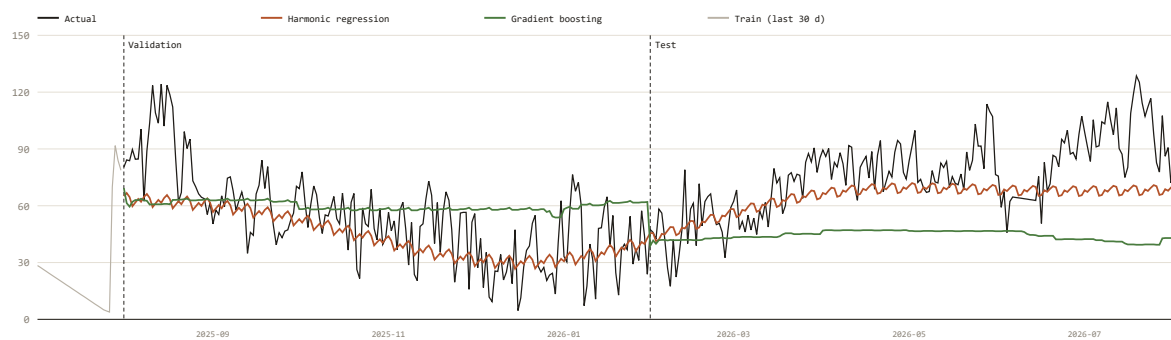
Computed over complete eight-hour windows only, so exceedances falling in incomplete windows are missed and the counts are a floor. The 2026 count covers 1 January to 1 August. Referenced in Section 4.

Figure A11. Forecast accuracy by model and evaluation period



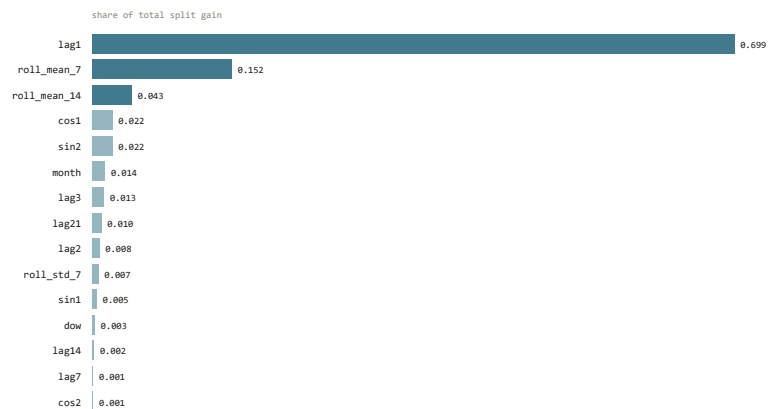
Lower is better. Values in $\mu\text{g}/\text{m}^3$; see Table A5. Referenced in Section 8.

Figure A12. Validation and test forecasts against actual values



Both models across both horizons on one axis, with the last thirty training days for context. The boosted forecast flattens within weeks of each origin; the harmonic forecast follows the season. Referenced in Section 8.

Figure A13. Gradient boosting feature importance



Share of total split gain across all boosting rounds. The one-day lag alone carries 0.699. Referenced in Section 8.

Table A1. Dataset metadata

FIELD	VALUE	NOTE
Location	Nîmes Gauzy (id 2162655)	Nîmes, Occitanie, France
Sensor / parameter	7774985 · o3 mass	Government monitoring network via OpenAQ
Units	µg/m ³	Mass concentration, not ppb
Raw resolution	Hourly, irregularly spaced	~80% of steps are 1 h; remainder 2 h or more
Modelling resolution	Daily mean	After resampling; see Section 5
Timezone	UTC stored, Europe/Paris local	Analysis in UTC; Figure A4 in local time
Extract span	2024-01-28 22:00 to 2026-08-01 17:00	917 calendar days, ends at date of writing
Hourly observations	14,398 raw · 14,039 usable	359 readings on 21 faulty days masked
Range / mean / SD	0 to 221.4 / 61.59 / 31.20	After masking; median 62.3, IQR 40.2 to 82.5
Hours above 180 / 240	5 / 0	Information threshold breached; alert threshold not
Consistency checks	0 duplicates, 0 negative, 0 above 500	445 exact zeros: 359 masked as fault, 86 retained
Source and licence	OpenAQ v3 API	Attribution required [6]

Table A2. Data structures used in the pipeline

STAGE	STRUCTURE	SHAPE / DTYPE	PURPOSE
API response	list of nested dicts	14,398 records	value, parameter, period, coverage, flagInfo
Raw frame	pandas DataFrame	14,398 rows, object	Dict-valued columns, unusable directly
Tidy frame	DataFrame [Date, Ozone]	datetime64[ns, UTC], float64	Persisted as the project CSV
Hourly series	Series, DatetimeIndex	14,039, irregular	Sorted, de-duplicated, fault days removed
Daily series	Series, freq="D"	917 × 1, float64	Regular grid; modelling unit
Gap mask	boolean Series	917 × 1, bool	Flags 49 interpolated days, 22 of them synthetic
MDA8 series	Series from hourly data	584 complete days	Regulatory statistic; Figure A10
Feature matrix	DataFrame (tree model)	896 × 15, float64	6 lags, 3 rolling, dow, month, 4 Fourier

Table A3. Missing days, the July 2025 fault, and hourly coverage

GAP START	CONSECUTIVE DAYS	CAUSE AND TREATMENT
2025-07-03	22	Instrument fault: 20 all-zero days plus the 5-6 July reporting gap. Masked, interpolated, flagged synthetic. A 21st fault day, 26 July, is a separate one-day gap
2026-06-07	7	No data reported. Time-interpolated
2024-03-12	6	No data reported. Time-interpolated
2024-03-19	4	No data reported. Time-interpolated
2024-02-28	3	No data reported. Time-interpolated
2024-08-15, 2025-07-05	2 each	Time-interpolated; the second adjoins the fault block
Plus the 26 July 2025 fault day and four single-day reporting gaps. Total 49 of 917 days (5.3%) carry no usable observation; 22 days (2.4%) lie inside the run longer than 14 days and are flagged synthetic.		
Hourly coverage on days with usable data		
2024	14.0 h/day	4,505 readings over 322 days
2025	16.7 h/day	5,675 readings over 340 days, after masking
2026	18.7 h/day	3,859 readings over 206 days

Table A4. Stationarity test results

SERIES	ADF STATISTIC	ADF CONCLUSION	KPSS STATISTIC	KPSS CONCLUSION
Level	-3.871	Rejects unit root at 1%	0.918	Rejects stationarity at 1%
First difference	-7.324	Rejects unit root at 1%	0.009	Does not reject stationarity
ADF with constant, lag order by AIC; critical values -3.44 (1%), -2.87 (5%), -2.57 (10%). KPSS level-stationary form, Bartlett window of 6 lags; critical values 0.739 (1%), 0.463 (5%), 0.347 (10%). The disagreement at the level is discussed in Section 6.				

Table A5. Forecast accuracy, validation and test periods ($\mu\text{g}/\text{m}^3$)

MODEL	VAL MAE	VAL RMSE	VAL MAPE	TEST MAE	TEST RMSE	TEST MAPE
Harmonic regression (Fourier K=2 + weekly)	15.19	19.88	36.3%	15.33	19.98	19.7%
Gradient boosting (44 rounds, depth 3)	18.57	24.07	65.7%	32.41	37.62	39.7%
Naive (last observed value)	30.09	34.87	106.5%	51.98	55.73	65.8%
Seasonal naive (7 days)	40.64	46.88	110.5%	37.35	43.21	46.8%

Train 551 days (28 Jan 2024 to 31 Jul 2025), validation 183 days (to 30 Jan 2026), test 183 days (to 1 Aug 2026). MAPE is greyed because it is unstable on this series: values near zero inflate it regardless of absolute accuracy. Referenced in Section 8.